



Evaluating the conservation risks of aggregate harvest management in a spatially-structured herring fishery



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ABSTRACT

Despite broad recognition of the potential for significant spatial complexity in morphological, behavioral, and life-history traits within many fish populations, fisheries are commonly managed across large spatial scales that aggregate interacting sub-populations into single management units. Such mis-match between the ecological and management scales may lead to a loss of spatial diversity and a restricted ability of populations to adapt and persist in the presence of changing environmental conditions. Developing harvest strategies for spatially complex fish populations therefore remains a major challenge. In this study, we evaluate whether managing spatially complex fish stocks as large-scale aggregates leads to greater conservation risks. We develop a closed-loop simulation model that represents a range of dispersal scenarios and includes imperfect management knowledge about the abundances and dynamics of interacting Pacific herring (*Clupea pallasii*) sub-populations, as well as weak management control of how exploitation is allocated among sub-populations. The latter is driven by the spatial dynamics of the fishing fleet as it seeks to optimize profitability. Simulated management outcomes did not always lead to increased risks under all scenarios of dispersal, fishery spatial dynamics, and management errors. Instead, these processes interacted to either mediate or intensify the impact of inappropriate management assumptions and stock assessment errors. Management strategies aimed directly at limiting exploitation risk consistently protected spatially complex populations in the presence of incorrect management assumptions about stock structure, high fishing power, and persistent stock assessment errors. Given the pervasiveness of these errors in fisheries, we recommend further evaluation of spatio-temporal refugia for tactical management of spatially complex fish populations.

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1. Introduction

Many exploited fishes exhibit a high degree of spatial variation in genetic, morphological, behavioral, and life-history traits within individual species and populations (Hilborn et al., 2003; Kerr et al., 2010; Schindler et al., 2010). Ignoring this spatial complexity when assessing and managing fisheries could erode spatial diversity, restrict species adaptation and persistence in the presence of environmental change, and reduce the success of harvest strategies (Cope and Punt, 2011; Hilborn et al., 2003). Simulations also indicate that unclear stock delineation

may have a greater impact on harvest strategy performance than uncertain abundance measurements and variation in mortality and recruitment (Punt and Donovan, 2007). One precautionary approach to dealing with such uncertainty may be to create small management areas that are unlikely to contain more than one population (Taylor et al., 2000). However, the opposite approach, most commonly used in fisheries, is to aggregate interacting sub-populations into single management units that are managed and exploited across large spatial scales (Cope and Punt, 2011).

Uncertainty about the ecological processes governing spatial structure within, and connectivity among, spatially complex populations has been part of the rationale for aggregate assessment and management of spatially diverse fisheries (Sinclair, 1988; McQuinn, 1997; Corten, 2002). However, a modern precautionary approach requires taking these uncertainties into account when developing fishery harvest strategies (Stephenson, 1999). Our current understanding about the risks posed by managing

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spatially complex populations as single, aggregate stocks depends on two common assumptions embedded within fishery models: (1) populations are discrete with no exchange of adult individuals between sub-populations (Frank and Brickman, 2000) and/or (2) fishing mortality is homogeneous across populations within a spatial complex (Wilson et al., 1999; Cope and Punt, 2011). Less research has been done on the risks of managing spatially complex populations that are connected by dispersal, which is an important mechanism affecting long-term persistence and productivity of spatially structured fish populations (Stephenson, 1999; Kerr et al., 2010; Schindler et al., 2010). Population connectivity, maintained by dispersal of individuals within metapopulations, is believed to create rescue effects that increase resilience to exploitation or other extinction risks (Dulvy et al., 2003). Rescue effects caused by dispersal are further believed to reduce risks of managing spatially complex fish populations as large-scale aggregates (Cope and Punt, 2011). Although exploitation typically occurs over broad spatial scales, fishing mortality is spread unevenly across spatially structured populations, typically concentrating on populations that present more profitable fishing opportunities (Rassweiler et al., 2012). Profitability can be related to distance from ports, time periods in which prices are high relative to fishing costs, or times of year, such as spawning seasons, when fish are easier to catch. Such spatial and temporal heterogeneity in profitability exposes some harvested populations to greater exploitation rates than others (Sanchirico and Wilen, 1999).

In this paper, we examine whether aggregate fisheries assessment and management leads to high conservation risks for spatially-structured populations. Specifically, we investigate how conservation risks are affected by interactions between adult dispersal within a metapopulation and fishery characteristics that determine how exploitation is distributed among sub-populations. A key distinction between our approach and related work on this topic is that we depict weak management control of how exploitation is allocated among sub-populations in addition to simulating imperfect management knowledge about the abundances and dynamics of the fish populations (e.g. Cope and Punt, 2011; Frank and Brickman, 2000). The former is driven by the spatial dynamics of the fishing fleet as it seeks to optimize profitability. We represent these assumptions in a closed-loop simulation model in which the management system propagates realistic errors in monitoring data, stock assessment models, and rules for determining annual total allowable catches (de la Mare, 1998; Punt, 2006; Cox and Kronlund, 2008). We simulate a management system that mimics the general characteristics of the Pacific herring (*Clupea pallasii*) fishery in the Strait of Georgia (SOG), British Columbia (B.C.), Canada (Fig. 1a).

1.1. The B.C. Pacific herring fishery system

Pacific herring are managed as five major and two minor stocks within B.C. However, genetic analyses indicate that four distinct

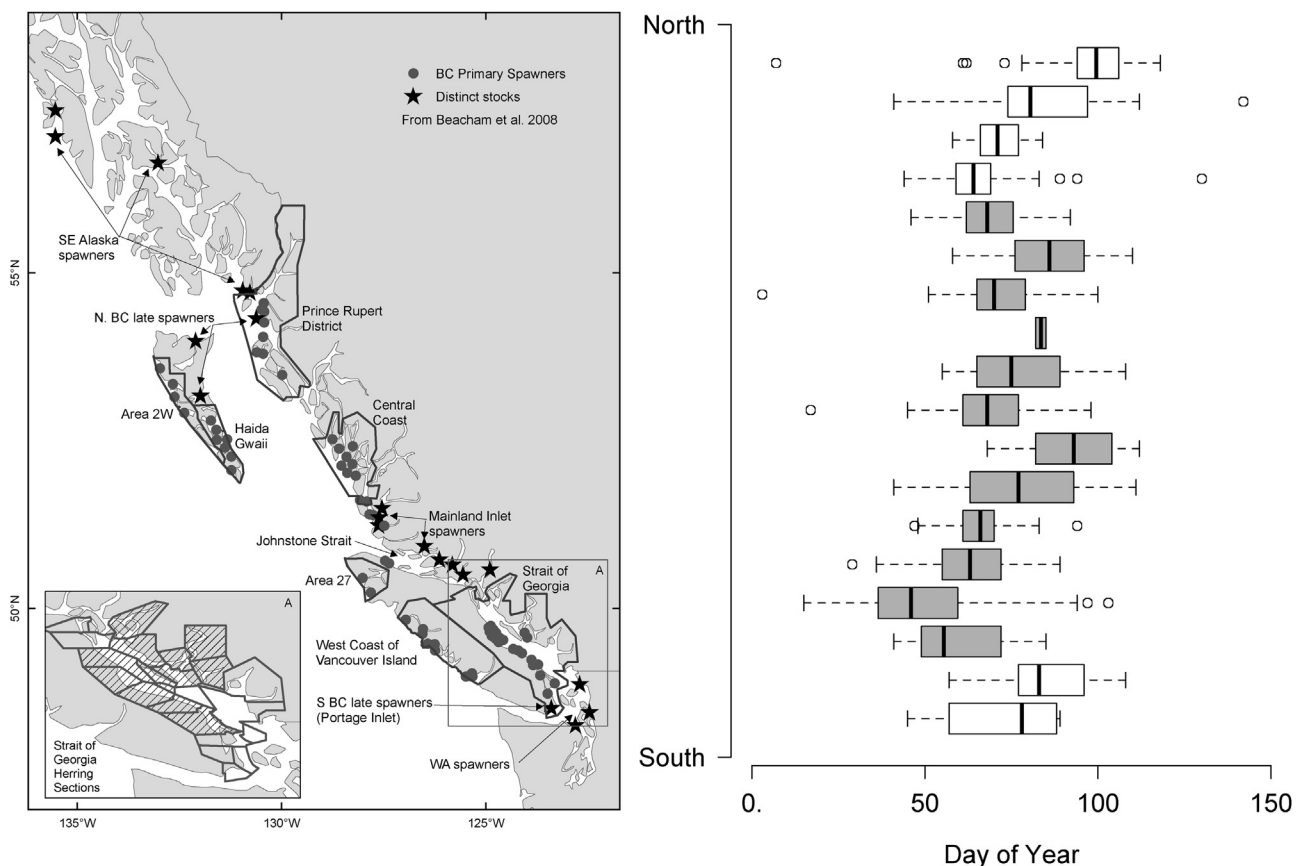


Fig. 1. (a) B.C. Pacific herring management areas, (grey boundary) and spawning locations of the B.C. primary spawners (grey filled circles) and the other genetically distinct stocks (black stars) identified by Beacham et al. (2008). Strait of Georgia (SOG) spawning sections shown on inset A; shaded (hatched) sections correspond to the 12 historically productive spawning "sections". (b) Empirical distributions for Pacific herring spawn timing for $P=18$ SOG spawning sections. Shaded boxplots indicate the sections and data used in this paper (1928–2006), and correspond to inset A. Boxplots in (b) summarize the median, 25th, and 75th percentiles of spawning dates, the whiskers correspond to the 5th and 95th percentiles, and the open circles are outliers.

herring stocks co-occur within the management boundaries: (1) B.C. primary spawners—a large ‘primary’ stock geographically dispersed throughout northern and southern B.C.; (2) Mainland Inlet spawners—extending from Johnstone Strait to the Central Coast; (3) Northern B.C. late spawners—found in Haida Gwaii and Prince Rupert; and (4) Southern B.C. late spawners—found in Portage Inlet (Beacham et al., 2008). Mainland Inlet spawners and the Northern and Southern B.C. late spawners all co-occur geographically with the B.C. primary spawners (Fig. 1a). Differences in timing of spawning are reported as the main mechanism isolating these stocks (Beacham et al., 2008). B.C. herring exhibit diversity in both morphology and spawn timing among spawning sites (Haegele and Schweigert, 1985, 1991), as well as diversity in dispersal behavior that complicates clear delineation of the true stock structure. For example, although spawning occurs in large bays and inlets throughout their distribution from B.C. to Korea, most herring undertake seasonal migrations from inshore spawning and nursery areas to offshore feeding grounds along the continental shelf, while a small resident portion remains close to spawning areas year round (Stevenson, 1946). In addition to the large-scale seasonal migrations, herring stray, abandon, and recolonize discrete spawning sites at a relatively fine spatial scale (Hay, 1985; Ware and Tovey, 2004). Spatial controls on fishing activity are not applied within the boundaries of the seven stock areas. Instead, the commercial fishery is timed to optimize herring roe quality as fish move into and out of spawning locations during the fishing season.

2. Methods

2.1. Overview of the simulation model

Our fishery simulation model has both operating model and management system components. The stochastic operating model simulates the spatial and temporal responses of spatially complex fish populations and fisheries to annual total allowable catch limits (TACs) set by the management system. A key feature of this simulation is that the management system assumes a single, homogeneously mixed fish population or aggregate stock, and is therefore unaware of how spatial diversity in spawn timing, recruitment productivity, and dispersal contribute to aggregate stock production. The aggregate stock assumption mimics the B.C. herring fishery management system where TACs are set within each of the stock areas.

We test the aggregate management system against a set of operating model scenarios that combine spatial complexity of the fish stock with numerical responses and harvesting power of the fishing fleet. Spatial complexity of fish populations is represented in a spatial population dynamics model as local variation in spawn timing, spawning habitat quality, and the following three alternate hypotheses about connectivity among spawning populations: (1) discrete (DIS) populations that are closed to immigration and emigration, and (2) metapopulations maintained by density-dependent dispersal (DDD) in which multiple herring stocks intermix via spawning habitat selection by adult fish, or (3) metapopulations maintained by density-independent dispersal (DID) in which adult fish disperse at random. The DIS scenarios are an extreme violation of the management system assumption of homogenous mixing because over-fishing of any stock has a direct, negative impact on total population production. The metapopulations scenarios most closely match the management assumption, although populations are not all equally productive. The consequences of incorrect management assumptions about stock structure depend on the interactions between herring spatial population dynamics and fishery spatial dynamics that determine

how TACs derived from the management system are taken from the populations. For example, in the absence of spatial controls on fishing effort, the *a priori* expectation is that the fishery takes the TAC by fishing in the highest-abundance spawning areas first, and sequentially moving on to fish lower-abundance areas as the season progresses. Such a fishing pattern may present high risks to some sub-populations because the unconstrained local catch may not be sustainable in the long-term. Conversely, mixing may replenish depleted sites and mitigate the impacts of heavy fishing on a local scale via a ‘rescue effect’ in which immigrants supplement subpopulation productivity (Smedbol and Wroblewski, 2002).

2.2. Operating model

2.2.1. Herring spatial population dynamics model

The herring spatial population dynamics model simulates $P = 12$ herring populations that represent a subset of historically productive spawning “sections” (spatial amalgamations of individual bays and beaches) within the SOG (Fig. 1). The model allows for either discrete populations or an interconnected mixture maintained by adult dispersal between the 12 spawning sections. We use a delay-difference model to provide a simple, yet biologically realistic description of each population biomass and numbers that accounts for biomass growth and lag effects in age-structured populations (Deriso, 1980; Schnute, 1985). The two main assumptions underlying our delay-difference model are: (1) all fish recruit to the fishery and contribute equally to the spawning stock at age-3 and (2) growth over ages follows the Ford–Brody growth function. Parameters of the weight-at-age function ($\alpha = 0.00004$ t and $\rho = 0.7067$) and weight-at-recruitment w_k (0.000091 t) were estimated based on herring fishery data from the SOG. We do not account for potential changes in natural mortality or growth in the SOG (Schweigert et al., 2010) as this was beyond the scope of this paper.

Because several dynamic processes occur within a single simulated year, we use the notation “–” to indicate biomass and numbers prior to fisheries and “+” to indicate their post-fishery values. The general delay-difference equations for the spatial model are of the form,

$$N_{p,t}^- = [SN_{p,t-1}^+ + R_{p,t}] (1 - E) + I_{p,t}^N \quad (1)$$

$$B_{p,t}^- = [S(\alpha N_{p,t-1}^+ + \rho B_{p,t-1}^+) + w_k R_{p,t}] (1 - E) + I_{p,t}^B \quad (2)$$

where $S = 0.63$ is the natural annual survival probability, $R_{p,t}$ is age-3 recruitment to spawning population p in year t , E is the proportional emigration rate, and $I_{p,t}^N = \gamma_{p,t} D_t^N$ and $I_{p,t}^B = \gamma_{p,t} D_t^B$ are the numbers and biomass, respectively, of fish immigrating to population p . For the discrete populations scenarios, the dispersal rate $E = 0$, which means that all emigration/immigration terms drop from the model. For mixed populations, $0 < E < 1$, and immigrants are calculated from population-specific immigration rates $\gamma_{p,t}$ (defined below) from numbers- and biomass-specific dispersal pools (D) given by,

$$D_t^N = E \sum_{p=1}^P SN_{p,t-1}^+ + R_{p,t} \quad (3)$$

$$D_t^B = E \sum_{p=1}^P S(\alpha N_{p,t-1}^+ + \rho B_{p,t-1}^+) + w_k R_{p,t} \quad (4)$$

such that all recruits redistribute spatially.

For discrete populations, the following unfished equilibrium conditions exist for the number of mature spawners and recruitment, respectively,

$$N_{p,0}^+ = \frac{B_{p,0}^+}{\bar{w}} \quad (5)$$

$$R_{p,0} = N_{p,0}^+(1 - S) \quad (6)$$

where $\bar{w} = (S\alpha + w_k(1 - S)) / (1 - \rho S)$ is the unfished equilibrium mean body weight.

Annual recruitment of age-3 fish to each sub-population is modeled using Beverton–Holt functions in which population productivity is assumed equivalent to local habitat quality (Walters et al., 2007). We represented habitat quality by sub-population-specific recruitment compensation factors K_p , which are the ratios of recruits-per-spawner at the origin (i.e., spawners $\rightarrow$ 0) to unfished equilibrium recruits-per-spawner (Goodyear, 1977). We use K_p values of 2.0, 2.0, 2.5, 2.5, 4.0, 5.0, 5.0, 4.0, 2.5, 2.5, 2.0, and 2.0 (which reflect “steepness” values between 0.33 and 0.56) to represent among-population variation in habitat quality between the 12 modeled herring sections. Although productivity and carrying capacity are typically considered independent in stock–recruitment models (Schnute and Kronlund, 2002), we apportioned biomass carrying capacities among sub-populations in proportion to productivity, i.e., $B_{p,0}^+ = B_{\cdot,0}^+ (K_p / \sum_p K_p)$. Annual recruitment for discrete populations is then given by

$$R_{p,t} = \begin{cases} R_{p,0} e^{\sigma_R \omega_{p,t} - \sigma_R^2/2} & 1 \leq t < k \\ \frac{K_p R_{p,0} B_{p,t-k+1}^+}{B_{p,0}^+ + (K_p - 1) B_{p,t-k+1}^+} e^{\sigma_R \omega_{p,t} - \sigma_R^2/2} & k \leq t \leq T \end{cases} \quad (7)$$

where k is the age at recruitment, T is the final simulation year, $\omega_{p,t} \sim N(0, 1)$ and $\sigma_R = 0.25$ is the arbitrarily chosen standard deviation in log-recruitment. We assume local density dependence in production because: (i) suitable herring spawning habitat is heterogeneously distributed along the B.C. coast, (ii) spawning activity is concentrated in sheltered bays and beaches (Hay et al., 2009), and (iii) spawning fish deposit eggs in layers during mass spawning events (Hay, 1985). Collectively these observations support the plausibility of processes such as habitat crowding in response to high population density.

The emigration rate in the migratory populations scenarios ($E=0.8$) implies that 80% of each population moves each year. This high rate of migration reflects the current management assumption that spawning herring populations are highly mixed. Density and population size are believed to be important factors that influence the choice and use of any given habitat (MacCall, 1990). We represent this process in the density-dependent dispersal (DDD) scenario in which emigrating spawners move among populations in response to expected differences in net reproductive output (Ruxton and Rohani, 1999). In particular, fish from the dispersal pools immigrate proportionately to sub-populations such that the logarithm of total recruitment over all sub-populations is maximized, i.e.:

$$\arg \max_{\gamma_{1,t}, \gamma_{2,t}, \dots, \gamma_{p,t}} \left(\log \left(\sum_{p=1}^P \frac{K_p R_{0,p} (B_{p,t}^- + \gamma_{p,t} D_t^B)}{B_{0,p}^+ + (K_p - 1) (B_{p,t}^- + \gamma_{p,t} D_t^B)} \right) \right) \quad (8)$$

where the immigrant proportions are subject to the probability constraints $\gamma_{p,t} \in [0, 1]$ and $\sum_{p=1}^P \gamma_{p,t} = 1$. This density-dependent

dispersal function generates a relatively homogeneous distribution of spawning biomass each year (Fig. 2a and c). We obtain equilibrium conditions for this DDD scenario by simulating the dynamic

model to convergence after initializing each population using Eqs. (5) and (6).

For populations connected by density-independent dispersal, the model is similar to density-dependent dispersal in that $E=0.8$, but the proportional immigration rates ($\gamma_{1,t}, \gamma_{2,t}, \dots, \gamma_{p,t}$) are drawn at random each year from a Uniform(0,1) distribution that is independent of population abundances and expected reproductive output (Fig. 2b and d). The values are standardized to sum to 1. Equilibrium conditions for this DID scenario are also obtained by simulating the dynamics to convergence.

Post-fishery spawning population numbers and biomass are updated after the fishery via,

$$N_{p,t}^+ = (1 - f_{p,t}) N_{p,t}^- \quad (9)$$

$$B_{p,t}^+ = (1 - f_{p,t}) B_{p,t}^- \quad (10)$$

where the harvest rates $f_{p,t}$ are dependent on the fishery dynamics (described below).

2.2.2. Fishery spatial dynamics model

The fishery dynamics model implements a quota-based fishery in which fishing season length is the only management restriction on how total catch is taken from the array of sub-populations over the fishing season. Daily catches from each sub-population are based on fish availability on individual spawning grounds and the fishery numerical response to that availability. Spatial and temporal variation in fish availability is aimed at mimicking features of B.C. herring. In particular, spatial variation in the seasonal timing of peak spawning is evident from spawn timing records collected in the SOG over the period 1928–2006 (Fig. 1b; Pacific herring database, Pacific Biological Station, Nanaimo B.C.). We represent this spatial variability in peak spawn timing by drawing a population-specific random mean day of spawning ($m_{p,t}$) each year from the spawn timing data for 12 historically important spawning and fishing sections that encapsulate a wide distribution of spawn dates in the SOG (Fig. 1b). Once the spawning season is underway, herring spawning events typically consist of early, peak, and late waves spread over a period of approximately 4–5 days (Hay, 1985). We use a normal probability density function to model this daily pattern of herring availability within each spawning season in a similar fashion to models of salmon abundance where fish are modeled through fisheries and onto spawning grounds (e.g., Flynn et al., 2006; Holt and Cox, 2008). The daily biomass of herring arriving at each spawning location is

$$b_{p,d,t} = B_{p,t}^- e^{-\frac{(d-m_{p,t})^2}{2l^2}} \left(\sum_k e^{-\frac{(k-m_{p,t})^2}{2l^2}} \right)^{-1} \quad (11)$$

where $d=1,2,\dots,D$ is day of the season and $b_{p,d,t}$ is the available biomass for sub-population p in year t if the fishery is open. The residence time $l=5$ days (assumed identical across populations) determines the spread of the arrival distribution over days.

Information on the spatial distribution and catch by the fishing fleet is not recorded for the B.C. herring fishery. Therefore, we were not able to build an empirical fleet dynamics model for generating local harvest rates on individual sub-populations. Instead, we assumed that fishing effort (i.e., boats) followed an ideal free distribution (IFD; Fretwell and Lucas, 1970; Gillis, 2003) such that the expected catch per boat is equalized among harvesting locations. Three primary assumptions underlying our particular IFD model are that (1) fleets have perfect knowledge of the spatial distribution of herring spawning stocks, (2) there are no barriers or travel costs to moving among herring spawning locations, and (3) there is no limit on the total amount of effort applied by the fleet (Gillis and Peterman, 1998; Cox and Walters, 2002). The first

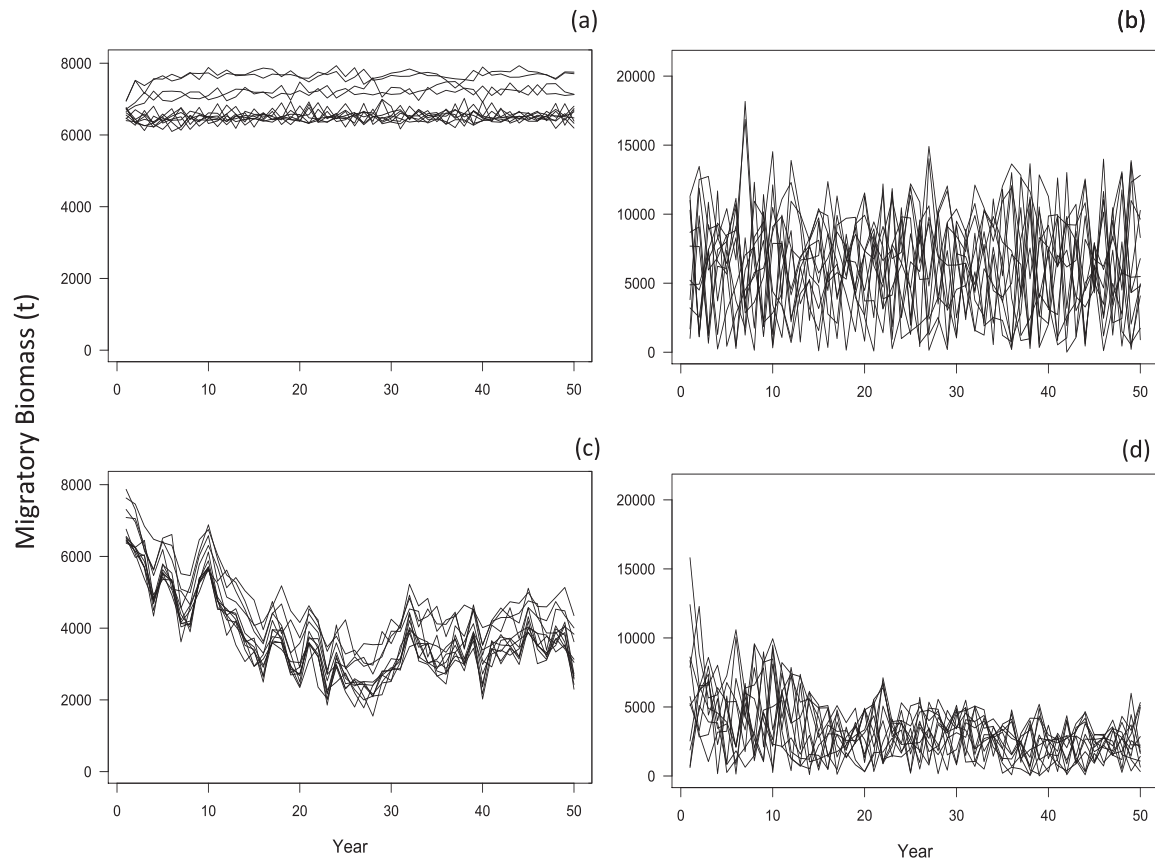


Fig. 2. Sub-population ($n=12$) biomass migration patterns for equilibrium unfished and baseline simulation settings for the density-dependent (a and c) and density-independent (b and d) mixed populations scenarios.

assumption is plausible for the SOG roe herring fishery because in-season estimates of spawning aggregation abundances (collected via acoustics) are publicized in near real-time to optimize harvested roe quality on short time-scales. The second assumption is reasonable because locations where herring aggregate prior to spawning within the SOG are also relatively accessible from numerous ports and harbors. Although the third assumption suggests that total effort could increase without limit over time on a given population, typical simulated daily effort on individual sites during the fishing season ranges from 1 to 82 boats, which appears reasonable given that there are approximately 440 vessels licensed to capture herring in B.C., and that in recent years most of the catch is taken out of the SOG in areas along the east coast of Vancouver Island from Comox to Nanaimo.

We used the fishing effort numerical response model derived by Cox and Walters (2002) because underlying process assumptions of a dynamic equilibrium between fish availability (e.g., movement on/off spawning grounds within a day), fishing effort, and catch provide biological meaning to the model parameters. Letting c_0 be the equalized IFD catch rate (tonnes per boat), q be the catchability coefficient (1/boat), and φ be the daily rate of movement on/off fishing grounds, the simulated catch on day d is

$$C_{p,d,t} = (b_{p,d,t} - 2c_0/q) U_{\max} \quad (12)$$

where $U_{\max} = 1 - e^{-\varphi}$ is the maximum possible daily harvest rate on each population (Cox and Walters, 2002) and the term inside the parentheses is the total biomass available to harvest on a given day. In keeping with the actual dynamics of the herring fishery, in which catches are monitored on the fishing grounds and validated within 24 h, the simulated cumulative seasonal catch is constrained

just below the quota recommendation. The final sub-population harvest rates for the whole fishing season are

$$f_{p,t} = \frac{1}{B_{p,t}^-} \sum_d C_{p,d,t} \quad (13)$$

These sub-population harvest rates may differ substantially from the aggregate target U_{MSY} set by the management system (defined below). The range of differences between realized harvest rates on each sub-population and the aggregate target depends on variation in available biomass (Eq. (11)) and the strength of harvesting response (Eq. (12)), where the latter depends on available biomass and the effort model parameters (c_0 , φ). We examined the sensitivity of conservation risks to these highly uncertain values (see below).

A single harvest rate trajectory is used to generate historical abundance dynamics and fishery data up to the first projection year $t = 21$ because the stock assessment model requires a time-series of data to estimate parameters and harvest control rule points. The assumed harvest trajectory for the first 20 years is the average harvest rate applied across populations and it remains fixed across simulations.

2.3. Management system model

We simulated a quota-based management system on the simulated herring stock. This system involves fishery-independent annual surveys of post-fishery spawning abundance, a stock assessment model to estimate aggregate spawning biomass and population status relative to estimated reference points, and a harvest control rule for computing the aggregate TAC each year from

the stock assessment estimates. We expand on each component below. Under all simulation scenarios, we assume perfect aggregate scale output control in which the fishery is closed when the annual TAC is reached.

2.3.1. Fishery-independent surveys

We simulated log-normally distributed annual fishery-independent surveys of post-fishery spawning biomass Y_t using

$$Y_t = B_{\cdot,t}^+ e^{\tau \delta_t - \tau^2/2} \quad (14)$$

where $B_{\cdot,t}^+$ is the total post-fishery spawning biomass summed over all sub-populations, $\delta_t \sim N(0, 1)$, and $\tau = \sqrt{\log(CV^2 + 1)}$ is the survey standard deviation and CV is the survey coefficient of variation (defined below). These survey biomass estimates represent the monitoring data used in the management system's stock assessment model.

2.3.2. Stock assessment model

For each projection year of the simulation (i.e., $t \geq 21$), we fit a spatially-aggregated delay-difference model to time-series of survey biomass estimates and catch. Results from this model are used to derive estimates of current spawning biomass ($\hat{B}_T$) (the “ $\hat{\cdot}$ ” symbol represents a quantity estimated or derived by the assessment model), forecast spawning biomass for the upcoming year ($\hat{B}_{T+1}$), and the current best estimate of unfished spawning biomass ($\hat{B}_{0,T}$). These quantities are needed in the harvest control rule computation of the TAC (defined below). We elected to use a full stock assessment model instead of a simpler estimation error approximation (e.g., Walters, 2004) because we wanted to explicitly represent the persistent stock assessment model errors arising from incorrect assumptions about stock structure and spatial population dynamics, both of which are potentially important errors in this type of management system.

The delay-difference stock assessment model assumes a single, homogeneously mixed spawning population, i.e.,

$$\hat{N}_t = \hat{S}_{t-1} \hat{N}_{t-1} + \hat{R}_t \quad (15)$$

$$\hat{B}_t = \hat{S}_{t-1} (\alpha \hat{N}_{t-1} + \rho \hat{B}_{t-1}) + w_k \hat{R}_t \quad (16)$$

where $\hat{S}_t = (1 - \hat{u}_t) e^{-M}$ is the estimated total annual survival rate and $\hat{u}_t = \text{TAC}_t / \hat{B}_t$ is the estimated exploitation rate. We provide the assessment with the true natural mortality rate. Annual recruitment is modeled using a constant median combined with annual recruitment anomalies ($\hat{\omega}_t$), i.e.,

$$\hat{R}_t = \begin{cases} \hat{R}_T & t = 1, t = T + 1 \\ \hat{R}_T e^{\hat{\omega}_t} & 1 < t \leq T \end{cases} \quad (17)$$

Parameters to be estimated by the assessment model are $\Theta_T = (\hat{R}_T, \{\hat{\omega}_t\}_{t=2}^T, q, \kappa^2)$ where t represents the year, and (q, κ^2) represent the survey catchability coefficient and total estimation error variance, respectively. The current estimate of unfished spawning biomass $\hat{B}_{0,T}$ is derived from the assessment estimate of unfished recruitment via:

$$\hat{B}_{0,T} = \frac{\hat{R}_T}{1 - S} \bar{w} \quad (18)$$

We used a mixed observation-process error maximum likelihood approach to estimate the values for the parameters of the delay-difference model. For the survey biomass index data, we first compute the observation error residual sum-of-squares Q_Y via

$$\hat{z}_t = \log(Y_t) - \log(\hat{B}_t - C_t) \quad (19)$$

$$\widehat{\log q} = \frac{1}{T} \sum_{t=1}^T \hat{z}_t \quad (20)$$

$$Q_Y = \sum_{t=1}^T (\hat{z}_t - \widehat{\log q})^2 \quad (21)$$

The process error residual sum-of-squares given the recruitment anomalies is

$$Q_R = \sum_{t=1}^T \hat{\omega}_t^2 \quad (22)$$

We then assign an equal proportion of the total error variance κ^2 to observation and process errors, leading to a total error variance estimate of the form:

$$\hat{\kappa}^2 = \frac{1}{4(T-1)} (Q_Y + Q_R) \quad (23)$$

Because the total error variance is unknown, the concentrated negative log-likelihood to be minimized is:

$$G = \frac{1}{4(T-1)} \log \hat{\kappa}^2 \quad (24)$$

For each projection year of the simulation, we conduct a stock assessment by minimizing G with respect to parameters Θ_T based on T years of data. On occasions where the assessment model does not converge, we generate biomass estimates and forecasts for year $T+1$ by projecting the most recent successful estimation using Eq. (16) and simulating catch for the intervening years. Convergence failure occurred most frequently in DID migratory populations that commonly realized rapid, one-way decline patterns of abundance.

2.3.3. Harvest control rule

The harvest control rule (HCR) specifies the TAC for year $T+1$ based on the estimated unfished biomass ($\hat{B}_{0,T}$), forecast exploitable biomass ($\hat{B}_{T+1}$), and an aggregate target exploitation rate (U_{MSY}). We used harvest control rule lower and upper control points $\hat{B}_{lower,T} = 0.2\hat{B}_{0,T}$ and $\hat{B}_{upper,T} = 0.4\hat{B}_{0,T}$, respectively, to reflect commonly used proxies for $0.5B_{MSY}$ and B_{MSY} (e.g. Shelton and Sinclair, 2008; Froese et al., 2010). This HCR is appropriate for the simulated populations, as the true aggregate B_{MSY} is $0.35\text{--}0.36B_0$ depending on the mixing scenario. The final parameter of the HCR is the reference harvest rate U_{MSY} , which is the harvest rate that, when applied equally across all operating model sub-populations, obtains the maximum sustainable yield. We provided the management system with the true aggregate scale U_{MSY} value (derived from Fig. 2a) because of well-known difficulties estimating production parameters from stock assessment models (Walters and Martell, 2004). Given this optimistic scenario, we also tested the impact of a less conservative policy using a biased U_{MSY} value. The annual TAC is computed via

$$\text{TAC}_{T+1} = \begin{cases} 0 & \hat{B}_{T+1} \leq \hat{B}_{lower,T} \\ U_{MSY} \left(\frac{\hat{B}_{T+1} - \hat{B}_{lower,T}}{\hat{B}_{upper,T} - \hat{B}_{lower,T}} \right) \hat{B}_{T+1} & \hat{B}_{lower,T} < \hat{B}_{T+1} < \hat{B}_{upper,T} \\ U_{MSY} \hat{B}_{T+1} & \hat{B}_{T+1} \geq \hat{B}_{upper,T} \end{cases} \quad (25)$$

Basing the HCR on estimated quantities rather than true operating model values propagates observation and process errors from surveys and stock assessments through to the management decision-making function, which more realistically represents closed-loop information feedbacks in real-world situations.

2.4. Simulation scenarios and sensitivity analyses

The combinations of operating model and management system models create many potential sensitivity analyses. We limit our attention to the following model parameters and levels (baseline model values are listed first): spatial distribution of habitat quality (normal, uniform), process error in recruitment ($\sigma_R = 0.25, 0.50$), observation error in the survey ($CV = 0.0, 0.25, 0.50$), fishing season length which specifies the days in which the fishery dynamics sub-model operates ($L = 40, 100$ days), fishery profit threshold ($c_0 = 15, 5$ t per boat), and fish availability to fisheries ($\varphi = 10, 1$ per day). The $CV = 0.0$ case represents a perfect information scenario in which the aggregate spawning biomass is known exactly by the management system assessment model. The fishery profit thresholds create fisheries that, respectively, can ($c_0 = 5$) and cannot ($c_0 = 15$) operate at low available biomass levels, which should be an important determinant of conservation risks. The fish availability parameters ($\varphi = 10, 1$) generate maximum local harvest rates $U_{\max} = 1.0, 0.4$, respectively, on any particular spawning population (Cox and Walters, 2002). Of all these parameters, only fishing season length (L) and the survey CV are under management control. We ran 100 Monte Carlo simulation trials for each of the 12 scenarios defined above.

2.5. Performance indicators

Performance of fishery management systems in closed-loop simulations is often judged by, for example, yield, biomass, or fishing mortality rates relative to stable equilibrium conditions. However, such metrics are difficult to interpret for the migratory populations scenarios because annual movement changes the relative productivity of the sub-populations as well as the aggregate production; therefore, equilibrium-based reference points such as B_{MSY} or F_{MSY} are of limited utility in deriving performance metrics. Instead, we use ratios of aggregate and sub-population biomass relative to their corresponding unfished starting conditions as a conservation metric. We refer to these biomass ratios as ‘depletion’ to be consistent with fisheries assessment terminology.

We summarize performance across the 100 simulation trials using median (over simulations) values of the following statistics:

- (1) DEP-AGG: average depletion of the aggregate population over projection years $t = 25$ to 50;
- (2) DEP-POPS: average depletion over sub-populations and projection years $t = 25$ to 50;
- (3) PROP < 20%: average proportion of sub-populations with depletion below 20% over projection years $t = 25$ to 50;
- (4) DEP-5TH: the 5th percentile of depletion across all sub-populations in projection years $t = 25$ to 50.

The DEP-AGG statistic provides a performance indicator from the perspective of the management system. DEP-AGG levels in the vicinity of 0.40 would reflect a management system that performs reasonably well given its goals of maintaining the aggregate population near the assumed most productive level. The DEP-POPS statistic indicates the average sub-population depletion, which, in a homogeneously mixed population, would also be near 0.40. However, for some spatial population dynamics scenarios, we expect DEP-POPS to deviate from DEP-AGG as the aggregate stock management assumption is violated. The PROP < 20% statistic and DEP-5TH represent indicators of conservation risks in the tails of the populations. Depletion values less than 20% of unfished levels are usually an indication of severe recruitment over-fishing risk, and the PROP < 20% statistic is a direct indicator of how many populations are depleted below that pre-determined level. The DEP-5TH

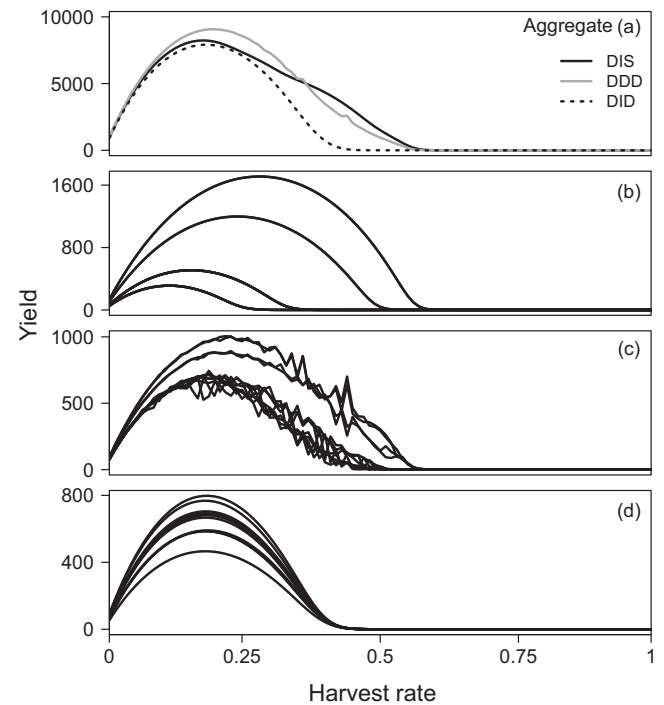


Fig. 3. Summary of equilibrium relationships between the annual exploitation rate and yield for (a) the population aggregate for each operating model connectivity scenario, and by sub-population for the (b) discrete (DIS), (c) density-dependent (DDD) and (d) density-independent (DID) mixed populations scenarios. These plots are generated by applying a constant exploitation rate over all sub-populations, in the absence of spatial effort dynamics by the fishery.

statistic is an indicator of the lowest depletion levels reached by individual sub-populations.

3. Results

Equilibrium properties of the spatial population dynamics model illustrate how average productivity differs among the discrete and mixed populations scenarios (Fig. 3). As expected, the individual yield curves for the sub-populations reflect the underlying distribution of habitat quality for the discrete populations (Fig. 3b). For DDD (Fig. 3c) or DID (Fig. 3d) metapopulations, biomass does not follow spatial variation in habitat quality exactly because dispersal re-distributes spawning biomass to less productive habitats, either increasing or decreasing aggregate biomass relative to the DIS scenarios.

3.1. Inseason dynamics

The results of one example simulation trial illustrate the average inseason behavior during the projection period for four combinations of management information and fishing (Figs. 4–6 DIS, DDD, and DID, respectively). Although the fish availability parameter in these scenarios allows potential daily exploitation rates up to $U_{\max} = 1.0$ on each sub-population, the high fishery profit threshold creates moderate-to-low total season harvest rates. Differences between DIS, DDD, and DID scenarios that were evident in the equilibrium properties are maintained in the stochastic projection: movement in the metapopulations scenarios that redistributes spawning biomass among sub-populations dampens the relative differences between sub-population biomass under both DDD and DID scenarios. Populations with spawn timing outside of the fishing season realize substantially lower harvest rates than populations for which spawn timing corresponds more closely with the fishing

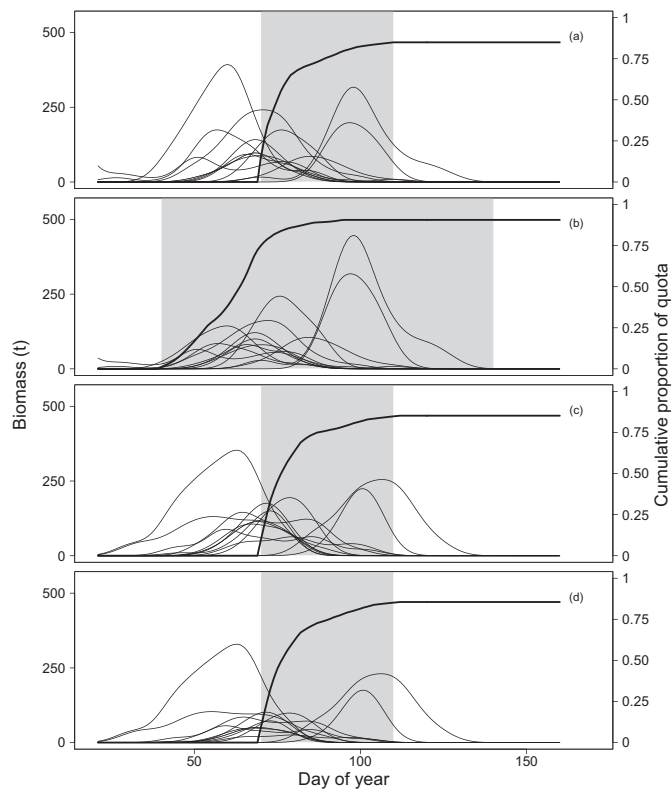


Fig. 4. Example results averaged over years 25–50 for one simulation under the discrete populations scenario. Plots show mean sub-population spawning biomass arriving on the spawning/fishing grounds and the cumulative daily proportion of quota during the fishing season (grey shaded area on each panel). Simulation conditions for each panel are: (a) baseline model settings ($c_0 = 15$, $L = 40$, $\varphi = 10$) with perfect information for the HCR; (b) a long fishing season ($L = 100$) with perfect information for the HCR; (c) baseline model settings using stock assessment estimates of biomass and low observation error; and (d) low profit threshold ($c_0 = 5$) and stock assessment estimates of biomass under low observation error.

season. Extending the fishing season to 100 days causes the fishery to preferentially exploit populations with higher biomass and early spawn timing (Figs. 4–6(b)). This results in earlier fishery closing times as total season harvest quotas get filled as indicated by the asymptote of the cumulative catch curves (Figs. 4–6).

The stock assessment model introduces substantial error into the management system and this impacts the performance indicators (Fig. 7). Biomass estimation errors exhibit a retrospective bias toward over-estimation during stock declines, and under-estimation during periods of stock increases. In general, this pattern holds across all three connectivity scenarios for the low-observation error scenarios, and is exacerbated in the high-observation error scenarios. However, the degree of bias and variability in the assessment model estimates of biomass varies between simulations and across scenarios, and can be significantly worse than the retrospective patterns presented in Fig. 7. In general, the effects of including assessment model errors in the simulation are higher exploitation rates and lower overall biomass across all discrete and metapopulation scenarios (6(c)).

Reducing the IFD catch rate from the baseline $c_0 = 15$ to $c_0 = 5$ t per boat in the DIS scenario results in the removal of most biomass available to the fishery due to high exploitation rates on sub-populations whose spawn timing overlaps with the fishing season. Under this DIS scenario, the early-spawning populations are maintained at relatively high biomass levels because spawn timing provides a temporal refuge from harvesting (Fig. 4(d)). Similar to the DIS scenario, DDD sub-populations show an intensification of harvesting on stocks that are available to the fishery as

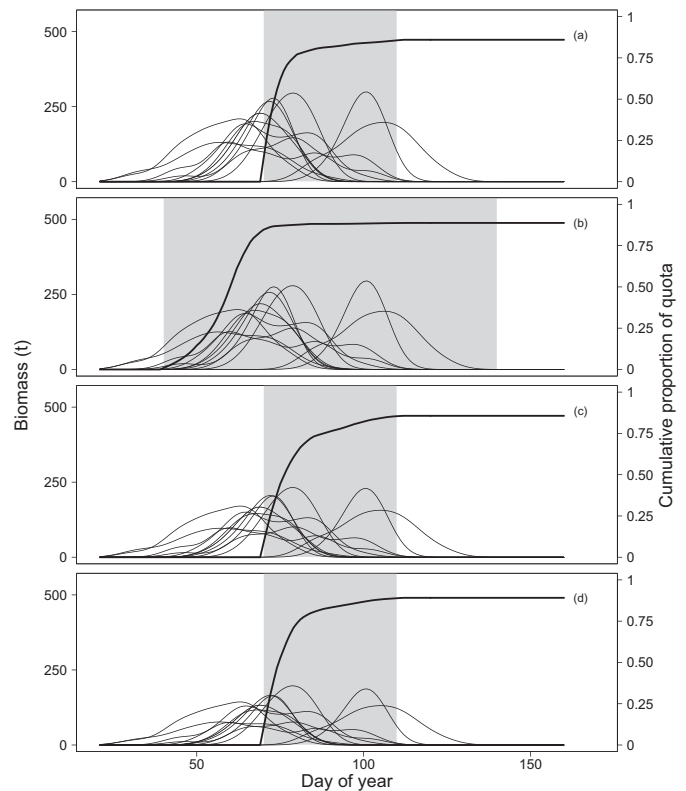


Fig. 5. Example results averaged over years 21–50 for one simulation under the metapopulations maintained by density-dependent dispersal. Panels and simulation conditions are identical to Fig. 3.

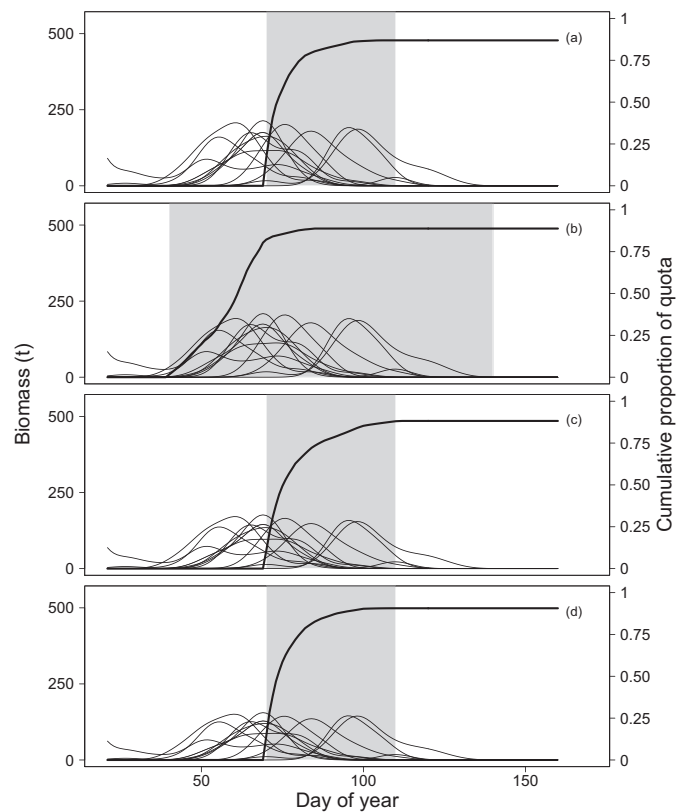


Fig. 6. Example results averaged over years 25–50 for one simulation under the metapopulations maintained by density-independent dispersal. Panels and simulation conditions are identical to Fig. 3.

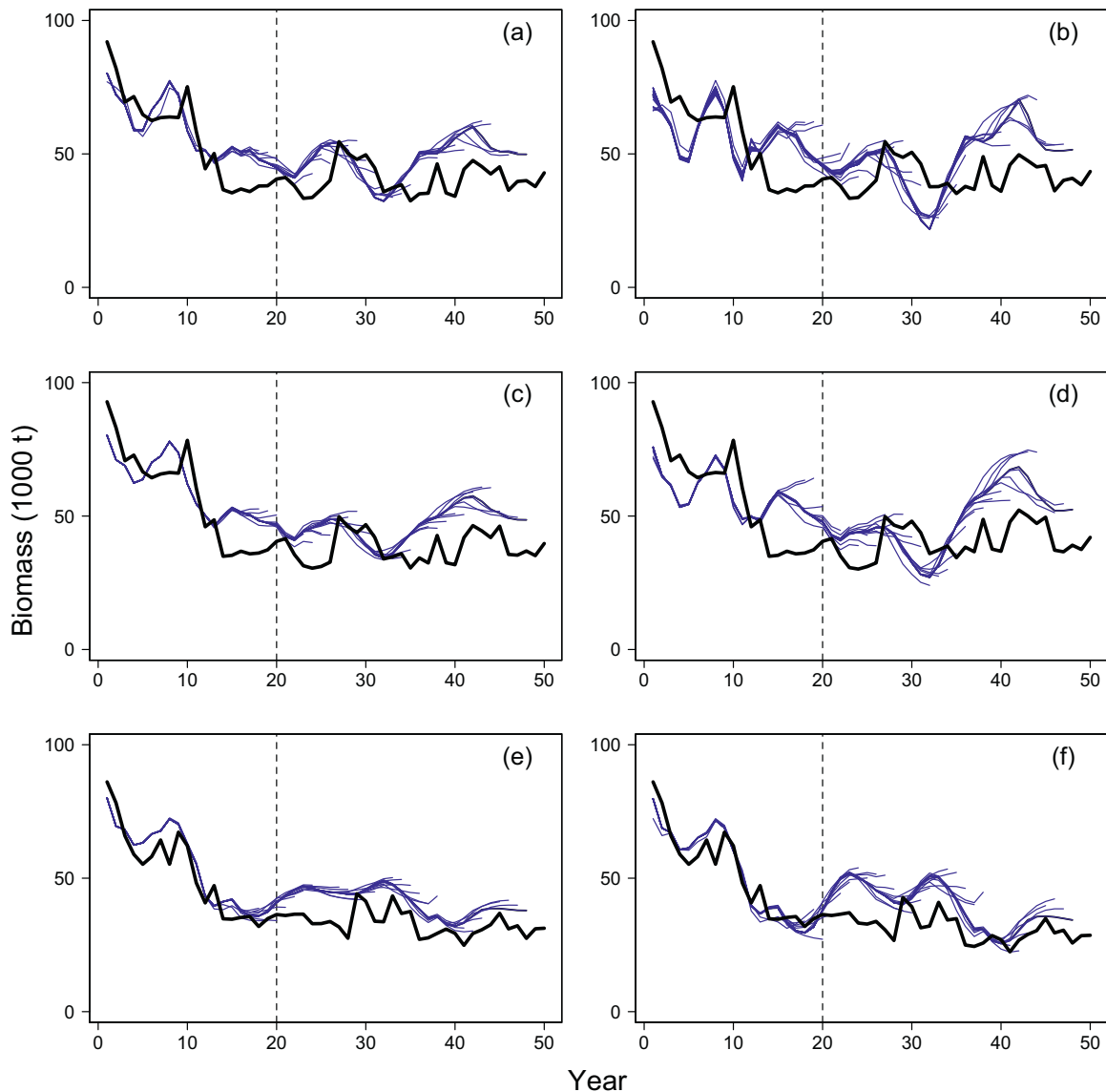


Fig. 7. Single simulation replicate example of retrospective biomass estimation patterns (blue lines) and ‘true’ biomass (thick black lines) for low (left column) and high (right column) observation errors under (a and b) discrete, (c and d) density-dependent, and (e and f) density-independent metapopulations scenarios. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the profitability requirement is lowered. However, in contrast to the DIS scenario, it is the late-spawning sub-populations that realize the lowest harvest impact (Fig. 5(d)). The late-spawning DDD sub-populations exhibit an additional, indirect impact of high exploitation rates, remaining at low levels of abundance despite not being harvested (moving onto spawning grounds after the catch levels off). This is due to reallocation of biomass to the most productive sites each year. In the DID scenario (Fig. 6(d)), the impact of increased harvest rates and lower biomass is shared across all sub-populations, which limits the effectiveness of seasonal harvest refuges.

The simulated management system reflects an ‘ideal’ in which catches are constrained to be below the quota and the management system is provided with the true aggregate scale U_{MSY} value. While we did not conduct full sensitivity analyses on these parameters, the asymptotic catches in all plots are below the point at which the quota constraint would be triggered (Figs. 4–6). This was unanticipated, particularly for the scenarios that increased fishing opportunity (specifically, the $L = 100$ day fishery and $c_0 = 5$).

Furthermore, adding a two-fold bias to U_{MSY} drives the asymptotic catch to approximately 50% of the quota (Fig. 8). This suggests that the spatial and temporal diversity in spawning behavior constrains the impact of inappropriate management due to either the stock assessment or the HCR.

3.2. Managing with perfect information

When the fishery is assessed and managed based on perfect information about aggregate biomass ($CV = 0$), aggregate depletion remains well above a level that would indicate a conservation concern (0.20) regardless of the population connectivity or fleet dynamics scenario (Table 1). However, performance measured at the sub-population level differs substantially from the aggregate for the DIS sub-populations, as reflected in DEP-POPS values below 40% for both the low fishery profitability and 100-day fishing season scenarios. DEP-POPS is maintained above 40% for both metapopulations scenarios regardless of fishery scenario. However, the PROP20% and DEP5th metrics highlight conservation

Table 1
Management procedure performance for the DIS, DDD, and DID connectivity scenarios for all operating model settings for habitat, season length, profitability and fish availability to the fishery. PROP<20% is the average proportion of populations with depletion below 20% and DEP-5TH is the 5th percentile of sub-population depletion. Bold values indicate levels of depletion consistent with an overfished state.

Survey error (CV)	Season length (L)	Profit Threshold (c_0)	Fish avail. (φ)	Depletion									PROP < 20%			DEP-5th		
				Aggregate (DEP-AGG)			Sub-populations (DEP-POP)											
				DIS	DDD	DID	DIS	DDD	DID	DIS	DDD	DID	DIS	DDD	DID	DIS	DDD	DID
Variable habitat/Low process error																		
0	40	15	10	0.46	0.46	0.49	0.37	0.45	0.48	0.06	0.00	0.11	0.26	0.26	0.16			
0	40	5	10	0.41	0.46	0.49	0.25	0.46	0.48	0.41	0.02	0.14	0.11	0.26	0.13			
0	100	15	10	0.42	0.45	0.48	0.30	0.45	0.48	0.12	0.01	0.12	0.23	0.24	0.16			
0	40	15	1	0.55	0.52	0.53	0.51	0.51	0.53	0.00	0.00	0.09	0.39	0.39	0.17			
0.25	40	15	10	0.35	0.31	0.35	0.31	0.30	0.35	0.14	0.14	0.21	0.23	0.19	0.12			
0.25	40	5	10	0.23	0.27	0.31	0.15	0.25	0.30	0.71	0.35	0.34	0.08	0.08	0.08			
0.25	100	15	10	0.24	0.27	0.32	0.24	0.25	0.31	0.28	0.40	0.25	0.20	0.16	0.11			
0.25	40	15	1	0.54	0.51	0.51	0.51	0.50	0.51	0.00	0.00	0.09	0.38	0.38	0.17			
0.50	40	15	10	0.35	0.31	0.35	0.31	0.31	0.35	0.13	0.15	0.20	0.23	0.19	0.12			
0.50	40	5	10	0.23	0.27	0.32	0.15	0.25	0.30	0.71	0.34	0.34	0.08	0.07	0.08			
0.50	100	15	10	0.26	0.27	0.33	0.24	0.26	0.31	0.28	0.40	0.25	0.20	0.16	0.11			
0.50	40	15	1	0.54	0.51	0.51	0.51	0.50	0.51	0.00	0.00	0.09	0.38	0.38	0.17			
Uniform habitat/Low process error																		
0	40	15	10	0.44	0.46	0.47	0.38	0.45	0.46	0.04	0.00	0.12	0.21	0.25	0.16			
0	40	5	10	0.41	0.46	0.47	0.28	0.45	0.45	0.29	0.03	0.16	0.10	0.24	0.12			
0	100	15	10	0.41	0.46	0.47	0.27	0.46	0.47	0.23	0.00	0.12	0.19	0.23	0.16			
0	40	15	1	0.57	0.55	0.55	0.55	0.55	0.55	0.00	0.00	0.08	0.41	0.42	0.18			
0.25	40	15	10	0.34	0.33	0.34	0.30	0.33	0.34	0.17	0.03	0.20	0.19	0.20	0.13			
0.25	40	5	10	0.22	0.26	0.27	0.16	0.26	0.27	0.65	0.36	0.45	0.07	0.08	0.08			
0.25	100	15	10	0.24	0.28	0.30	0.20	0.26	0.28	0.64	0.25	0.26	0.18	0.18	0.11			
0.25	40	15	1	0.57	0.55	0.55	0.55	0.54	0.54	0.00	0.00	0.08	0.40	0.42	0.18			
0.50	40	15	10	0.34	0.33	0.35	0.30	0.33	0.34	0.17	0.03	0.20	0.19	0.20	0.13			
0.50	40	5	10	0.22	0.26	0.27	0.17	0.26	0.27	0.65	0.36	0.44	0.07	0.08	0.08			
0.50	100	15	10	0.25	0.29	0.30	0.20	0.27	0.29	0.64	0.24	0.26	0.18	0.18	0.11			
0.50	40	15	1	0.57	0.55	0.55	0.55	0.55	0.54	0.00	0.00	0.08	0.41	0.42	0.18			
Variable habitat/High process error																		
0	40	15	10	0.47	0.47	0.49	0.38	0.45	0.48	0.06	0.01	0.12	0.24	0.23	0.15			
0	40	5	10	0.42	0.47	0.49	0.27	0.46	0.48	0.39	0.04	0.16	0.10	0.23	0.13			
0	100	15	10	0.44	0.46	0.48	0.33	0.45	0.50	0.12	0.02	0.13	0.21	0.22	0.15			
0	40	15	1	0.55	0.52	0.47	0.51	0.52	0.49	0.00	0.00	0.20	0.34	0.33	0.12			
0.25	40	15	10	0.37	0.34	0.36	0.33	0.32	0.36	0.15	0.17	0.22	0.22	0.18	0.12			
0.25	40	5	10	0.23	0.25	0.29	0.15	0.24	0.27	0.71	0.46	0.42	0.08	0.07	0.07			
0.25	100	15	10	0.27	0.28	0.32	0.25	0.26	0.32	0.29	0.44	0.29	0.19	0.13	0.10			
0.25	40	15	1	0.54	0.51	0.51	0.50	0.50	0.51	0.00	0.00	0.10	0.32	0.32	0.16			
0.50	40	15	10	0.37	0.34	0.36	0.33	0.33	0.36	0.14	0.17	0.22	0.22	0.18	0.12			
0.50	40	5	10	0.24	0.25	0.29	0.15	0.24	0.28	0.70	0.41	0.42	0.08	0.07	0.07			
0.50	100	15	10	0.26	0.28	0.30	0.25	0.27	0.28	0.29	0.43	0.32	0.19	0.14	0.09			
0.50	40	15	1	0.54	0.51	0.51	0.50	0.51	0.51	0.00	0.00	0.10	0.32	0.32	0.16			

concerns for both the DIS and DID scenarios even where management has the luxury of perfect information (Table 1). For example, under the low-profitability scenario, the PROP20% metric indicates that greater than 40% of DIS sub-populations were depleted below 20% of their stock-specific unfished biomass levels. Under the same scenario, DEP5th=0.11 further indicates that at least 5% of populations were depleted to 11% of their stock-specific unfished levels in any particular year.

3.3. Managing with imperfect information

With the exception of the strongly limited fish availability scenario ($\varphi = 1$), conservation performance degrades for all population scenarios when observation errors and stock assessment models are added to the management system. For example, both DEP-AGG and DEP-POPS are consistently below 40% for all population scenarios in the low observation error case ($CV=0.25$). PROP20% increased by a minimum of 10% across all scenarios, and by as much as 30% for DIS under the low-profitability case and 38% for DDD in the 100-day fishing season (Table 1). Similarly, DEP5th decreased across all scenarios, reaching a minimum of 8% for all connectivity scenarios in the low-profitability case. There is little

difference in simulation outcomes between the low and high survey observation error ($CV=0.50$) cases, presumably because stock assessment model performance was already poor for the low error case (Fig. 6, Table 1).

Simulation outcomes were generally insensitive to alternative scenarios of habitat distribution and recruitment variability: depletion measured at the sub-population level tends to be lower than the aggregate measure for the discrete populations, indicators of the poorest performance at the sub-population level reflect conservation concerns for all connectivity scenarios under 'realistic' management models (i.e. not perfect information), and stock assessment model error, fishery profitability, and fish availability are the key determinants of conservation risks (Table 1). However, while the absolute differences are small, depletion measured for the DIS populations with uniform habitat distribution tend to be lower than the baseline simulations. This difference is not observed for either the DDD or DID scenario.

4. Discussion

We questioned whether managing spatially complex fish stocks, such as Pacific herring in British Columbia, as large-scale

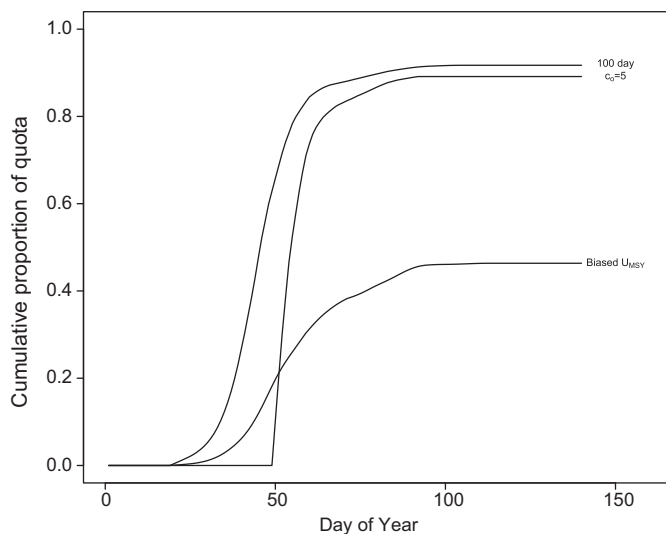


Fig. 8. Example of the cumulative daily proportion of quota during the fishing season for metapopulations maintained by density-dependent dispersal. Results are presented for combinations of management and fishery dynamics that maximize fishing opportunities and effort (long fishing season ($L = 100$), short fishing season ($L = 40$) with low profit threshold ($c_0 = 5$), and long fishing season ($L = 100$) with a biased U_{MSY} in the HCR), and reflect constraints on catch that arise due to spatial and temporal interactions of the fish and the fishery.

aggregates leads to greater conservation risks. Simulated management outcomes did not always lead to increased risks under all scenarios of dispersal, fishery spatial dynamics, and management errors. Instead, these processes interacted to either mediate or intensify the impact of inappropriate management assumptions and stock assessment errors. Management performance was strongly influenced by the fishing fleet/harvesting dynamics, regardless of population spatial complexity or stock assessment errors. Specifically, conservation risks were mainly influenced by the fishery profitability threshold and fishing opportunity, which represent spatial exposure (profitability) and temporal exposure (spatial and temporal overlap between the harvested populations and the fishery). Among-population variation in both timing and spatial distribution of biomass on the spawning/fishing grounds controlled the daily availability of each population to the fishery. In scenarios where the fishing effort distribution is driven by a high profit threshold, low levels of available biomass attracted little fishing effort. This IFD dynamic is compensatory, creating spatial refugia from fishing at the sub-population level (for low abundance populations) and also temporal refugia within a fishing season (for high abundance populations) because daily availability dynamics expose only a fraction of each sub-population to fishing at any moment in time. Such refugia are strongest for the high profit threshold and short fishing season simulations, which achieved biologically conservative outcomes because they combine spatial and temporal refuges across the broadest range of populations. In contrast, processes that intensify overlaps between the spatial distribution of fishing effort and the harvested populations promote high levels of fishing effort on a local scale, and contribute to depletion of the aggregate population. Examples of overlap-intensifying processes include: (1) longer fishing seasons that permit harvesting on all spawn timing groups both within and among sub-populations, (2) low profit thresholds that promote harvesting of small population abundances, and (3) strong mixing between sub-populations.

The structure of our simulation model omits some important dynamics and processes that could affect the results. Foremost is that the model is spatially implicit and therefore the harvest dynamics component does not account for the costs associated with

travel time and distance between fishing locations, which could influence the distribution of the fishery such that locations that are close to each other and to fishing ports realize higher exploitation rates than more distant locations (Hilborn and Walters, 1987). However, because the modeled region of the SOG is small relative to the dispersive capacity of the B.C. herring fleet, travel costs are justifiably ignored in this study. Another important feature of the model structure is that the population dynamics do not generate patterns of abandonment and re-colonization of spawning locations arising from local extirpation or natural changes in patterns of dispersal, as is expected under some models of spatial population dynamics (Hanski, 1998). This means that permanent loss of spatial diversity is not represented by the model. Spatial diversity is instead 'eroded' as the sub-populations are depleted to low levels of biomass under exploitation rates that vary significantly within and between years.

The spatial population dynamics model assumed that the M was constant, which is contrary to current evidence that M is changing rapidly for some herring stocks. In addition, we provided the management system the true M value. Estimated changes in M could reflect actual mortality patterns, perhaps in response to recovering marine mammal populations (e.g., humpback whales, seals, sea lions). However, apparent changes in M could also result from changes in spatial population structure and dispersal. Although we did not account for survival and fecundity costs of dispersal, theoretical studies suggest that these costs could be high (Delgado et al., 2014). Adaptive dispersal with movement costs could be a valuable addition to the spatial population dynamics model because it would provide a more realistic representation of density-dependent dispersal. Our simulations showing relatively little sensitivity of density-dependent dynamics to inappropriate management scales could change if dispersal costs were taken into account because these scenarios had the strongest movements.

The simulated management procedures understate uncertainty because the stock assessment model is provided starting estimates of average recruitment each year. This is necessary to achieve stable model performance across the suites of operating models we use in our analysis. In addition, our fishing effort dynamics model is based on parameters and assumptions that may not apply to the B.C. herring fishery. This limitation arises largely because spatial fishing effort and catch information are not collected for this fishery. However, future work could be informed by interviews with managers, local First Nations, commercial harvesters, and fishery scientists to develop a more realistic model of the fishery spatial dynamics. A similar limitation exists for the population simulator, which is based on generalizations of herring dynamics and makes no attempt to reproduce historical trends in biomass and exploitation specifically for B.C. herring. However, the objective of our study was to explore a general fishery management system that ignores spatial complexity in both the fish populations and fishery, and to identify points at which intervention achieves large changes in management performance. These general management implications are not expected to change with a more specific model.

Despite these limitations, our simulation results support previous empirical evidence that rescue effects do not always apply in exploited metapopulations; that is, populations with high connectivity are not necessarily more resilient to exploitation than discrete stocks (Orensanz et al., 1998). Simulated outcomes for connected populations were similar to, or slightly worse than, the discrete populations under most realistic management scenarios (i.e., those including assessment model error). Lower resilience occurred for the connectivity scenarios because both density-dependent and density-independent dispersal processes spread the impacts of fishing across populations that vary in productivity. In contrast, the impact of fishing is localized in the discrete populations scenarios, where less productive populations benefit from the compensatory

IFD response. The differences between outcomes for discrete and connected populations indicate that spatial and temporal dynamics of both the fish population and the fishery define the exploited stock. Therefore, as we illustrate, the impact of fishing on spatially complex populations cannot be inferred simply from the characteristics of the population or the scale of management. Depending on the dynamics of population migration and the fishery, high-connectivity populations may be as vulnerable to overfishing as discrete populations.

We showed that the combined spatial dynamics of fish populations and fisheries have powerful implications for fisheries management. In particular, aspects of spatial structure that appear most important for management (i.e., processes governing the overlap between populations and fisheries) may not be reflected in standard measures of stock structure such as genetic diversity. This issue is acknowledged by researchers in stock identification (Cadrin et al., 2005) who note that the definition of a “fish stock” is fluid, and depends on the reason for defining a stock in the first place (Carvalho and Hauser, 1994). Our results support the “harvest stock” concept that identifies stocks based on accessibility of fish to the fishery (Gauldie, 1988). This implies that the spatial and temporal overlap between fish and fisheries is a fundamentally important characteristic of spatially complex populations. Furthermore, economic aspects of the fishery appear to be equally important for defining a “stock” because they determine the profit incentives and capacity for harvesting. Fish price is relatively easy to obtain. However, economic evaluation requires additional information that is not routinely collected for many fish populations, including B.C. herring (e.g. vessel searching activity, catch locations, catch rate, fishing costs). If such information was available, the relationship between the spatial distribution of fishing effort and profit could be used to identify (or preserve) potential refuges from harvest within seasons and across space. Management tactics involving such spatio-temporal harvest refugia were historically successful for Pacific salmon and Australian rock lobster (*Panulirus cygnus*) (Walters and Pearse, 1996).

Incentives driving the fishery spatial distribution and controls placed on the spatial distribution and magnitude of effort are two important points of leverage in fishery management systems (e.g. Walters and Pearse, 1996; Sanchirico and Wilen, 2005). In our simulations, temporal controls on the number of days the fishery is permitted to harvest any particular sub-population had a significant effect on conservation risks by controlling the sub-population level impact of the fishery. Such temporal management tactics are most notably used for in-season management of salmon fisheries where (i) fisheries are managed to achieve population-specific spawning escapement and exploitation rate goals (Martell et al., 2008); (ii) fish availability and abundance vary dramatically throughout the fishing season (Walters, 1997), and (iii) exploitation proceeds despite considerably high uncertainty about population-specific abundances. For quota-based fisheries, such as Atlantic herring (*Clupea harengus*), temporal management tactics are used in-season to protect individual spawning populations within an aggregate (Stephenson et al., 1999; Claytor, 2000). We showed that a simple control on total fishing season length consistently limits conservation risks for spatially complex populations despite clearly incorrect management assumptions about the spatial dynamics of fish populations and the fishery. Nevertheless, further research on spatial population dynamics and harvester behavior would improve our ability to use temporal management tactics wisely to achieve a balance between conservation and exploitation. In particular, it is important to better understand how changes in fishing power and economic factors could partially offset the benefits for fish populations.

Stock assessment estimates of abundance are often highly unreliable because the fundamental assumptions underlying the models

are flawed (common examples include stationary production relationships and catch per unit effort proportional to abundance) (Hilborn and Walters, 1992). In addition, different stock assessment methods can differ dramatically in their ability to estimate reference points such as average recruitment or B_0 (Haltuch et al., 2008). Our results indicate that the effect of stock assessment errors on management performance depends on the degree of spatio-temporal overlap between the populations and the fishery. Removing all spatial and temporal controls on fishing (i.e., 100-day fishing season scenario) enabled the fishery to capture the entire TAC by fishing on all populations. Therefore, when the TAC was set to an inappropriate level, such as when stock assessment estimates were biased as a result of observation errors and model mis-specification, the system moved toward a persistently over-fished state. Imposing a broad temporal control on the fishing season restricted harvesting to a subset of more productive populations, improving conservation and fishery outcomes, even in the presence of biased stock assessments. In conclusion, given the pervasiveness of stock assessment modeling challenges, our analysis suggests that recognizing and managing fishing operations to account for potential spatial population complexity may increase the degree of precaution in quota-based management systems.

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